Microsoft Elevate

CAPSTONE PROJECT

CROP PRODUCTION INSIGHTS - INDIA

Presented By

STUDENT NAME : SRIRAMOJU TEJASWINI
COLLEGE NAME :Swarna Bharathi Institute of science and Technology, Khammam
DEPARTMENT : COMPUTER SCIENCE AND ENGINEERING
EMAIL ID : tejaswisrinivas24@gmail.com

OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

Agricultural production in India is influenced by several factors such as seasonal changes, crop variety, and regional differences. Large volumes of agricultural data are generated every year, but identifying meaningful patterns and trends from this data is challenging. Without proper analysis, it becomes difficult to understand which regions contribute the most to production, how seasonal variations affect crop output, and how production trends change over time. This creates a gap in gaining clear insights into crop production patterns across different states and seasons in India.

PROPOSED SOLUTION:

- The proposed system aims to analyze crop production data across different regions and seasons in India to identify meaningful patterns and trends. The objective is to organize and interpret large agricultural datasets in a structured manner so that production insights can be easily understood and explored. The solution will consist of the following components.
- **Data Collection:**
  Collect historical agricultural data containing information such as state name, district name, crop type, season, crop year, cultivated area, and production levels. This data provides the foundation for understanding agricultural performance across different regions and time periods.
- **Data Preprocessing:**
  Clean and prepare the dataset by handling missing values, removing inconsistencies, and ensuring that all fields are properly formatted. Data preprocessing also involves organizing the dataset into a structured format so that it can be easily analyzed and visualized.

- **Data Analysis:**
  Perform analytical operations to study production patterns across different states, crops, and seasons. This step helps in identifying trends, variations in production, and the contribution of different regions to the overall agricultural output.
- **Visualization and Dashboard Development:**
  Create interactive visualizations that represent key agricultural insights such as total production, seasonal distribution, and production trends over the years. The dashboard allows users to explore the data visually and gain a clear understanding of crop production patterns.
- **Deployment and User Interaction:**
  Present the final dashboard in a structured and user-friendly format where users can interact with the visualizations. The dashboard enables filtering and exploration of agricultural data to better understand production insights across different dimensions such as crop type, season, and region

SYSTEM APPROACH:

- **Hardware Requirements**
- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free disk space
- **Software Requirements**
- Operating System: Windows 10 or above
- Tool Used: Microsoft Power BI
- **Data Source**
- Crop Production dataset in CSV format
- The dataset contains attributes such as State Name, District Name, Crop, Season, Crop Year, Area, and Production.
- This data is used to analyze agricultural production trends and patterns across different regions.

Libraries / Tools Required to Build the Model

- **Power BI Desktop**
  Microsoft Power BI Desktop is the primary tool used to build the analytical dashboard. It allows importing datasets, creating data models, and designing interactive visualizations. Using its drag-and-drop interface, various charts such as line charts, KPI cards, donut charts, and bar charts were created to represent crop production insights clearly.

- **Power Query Editor**
  Power Query Editor is used for data transformation and preprocessing. It helps in cleaning the dataset by removing duplicate records, correcting data types, and handling missing or inconsistent values. This step ensures that the dataset is structured properly before performing analysis.

- **DAX (Data Analysis Expressions)**
  DAX is used to create calculated measures and perform advanced data calculations. Measures such as **total production, average production, and total cultivated area** can be defined using DAX formulas. These calculations help in generating meaningful insights and improving the analytical capability of the dashboard.

ALGORITHM & DEPLOYMENT:

Algorithm Selection

- The project uses a **data analytics and visualization approach** to analyze crop production patterns across different states, seasons, and years. Instead of predictive machine learning models, the system focuses on analyzing historical agricultural data to identify trends and patterns.

Data Input

- The algorithm processes agricultural data with the following features:
- State Name
- District Name
- Crop
- Season
- Crop Year
- Area
- Production
- These attributes help in understanding crop production variations across regions and time periods.

Processing Process

The dataset goes through several processing steps before analysis:

•Import the dataset into Microsoft Power BI

•Clean and transform the data using Power Query

•Create calculated measures using DAX

•Organize the data for analysis and visualization

Insight Generation

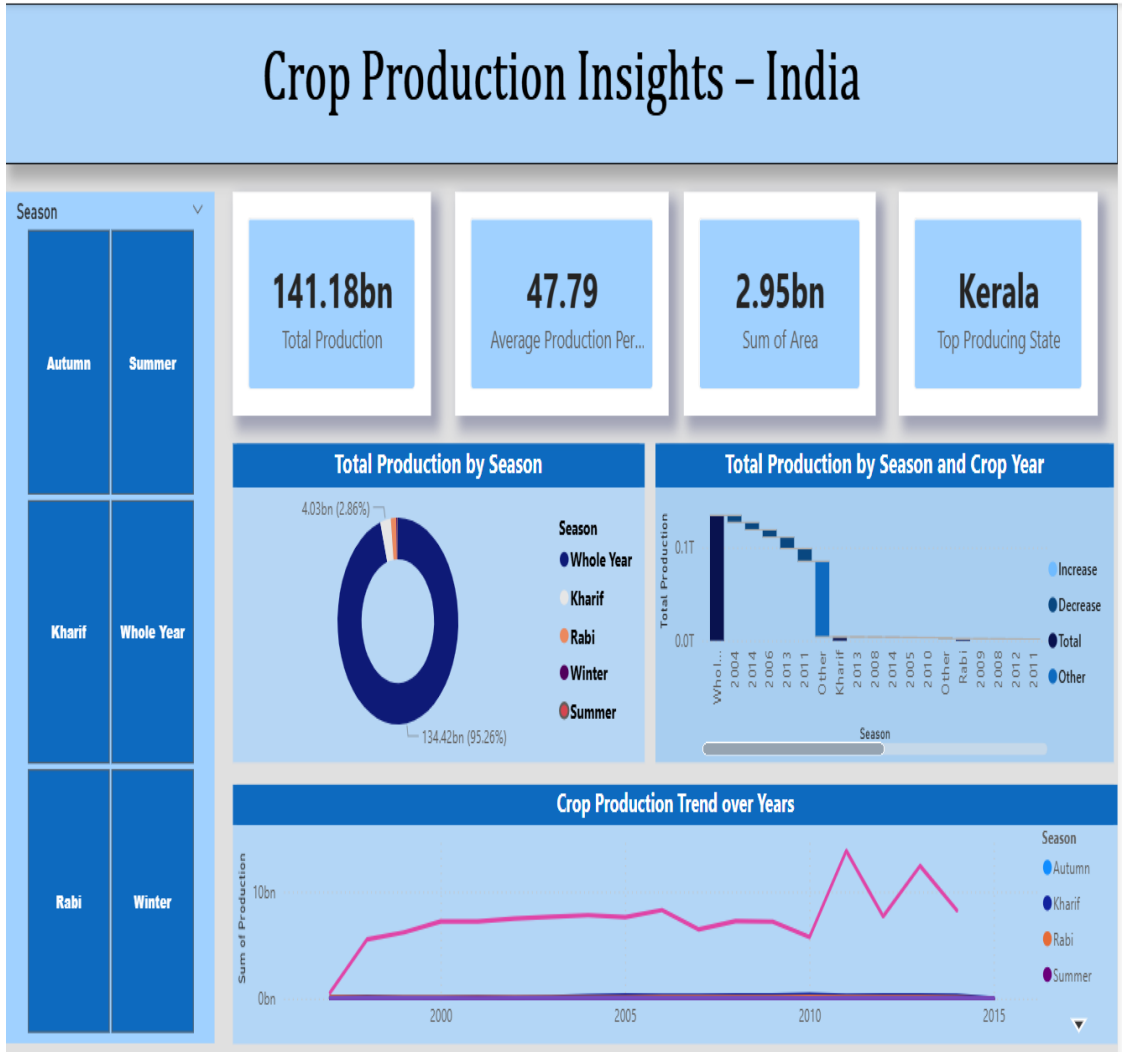
Visualizations such as KPI cards, line charts, donut charts, and bar charts are used to display production trends, seasonal distribution, and top producing crops or states.

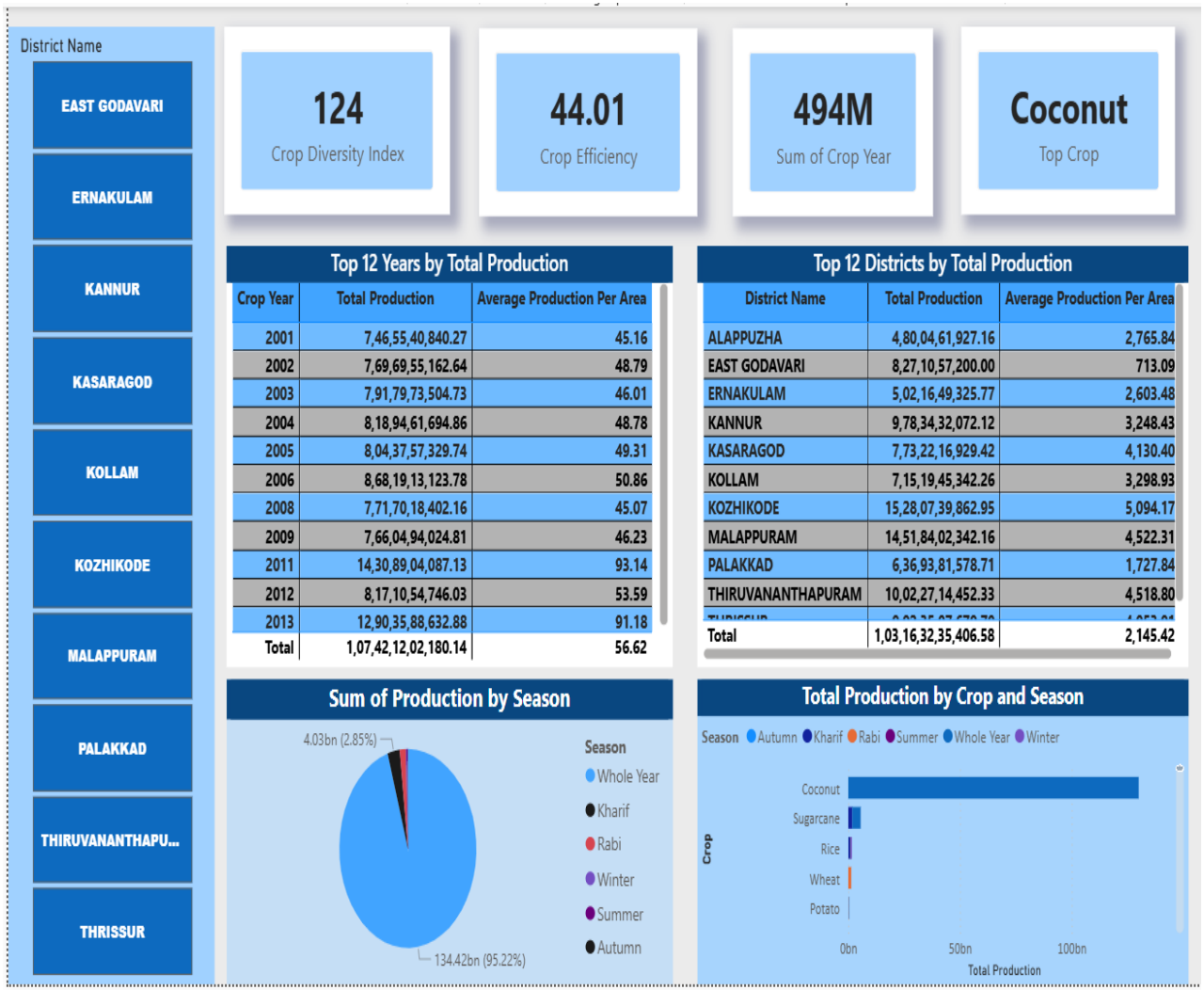
Deployment

Deployment Process

The final dashboard is developed and presented using Microsoft Power BI. The dashboard allows users to interact with the data using filters and slicers to explore crop production insights across different regions, seasons, and years.

RESULT:





Crop Year	Total Production	Average Production Per Area
2001	7,46,55,40,840.27	45.16
2002	7,69,69,55,162.64	48.79
2003	7,91,79,73,504.73	46.01
2004	8,18,94,61,694.86	48.78
2005	8,04,37,57,329.74	49.31
2006	8,68,19,13,123.78	50.86
2008	7,71,70,18,402.16	45.07
2009	7,66,04,94,024.81	46.23
2011	14,30,89,04,087.13	93.14
2012	8,17,10,54,746.03	53.59
2013	12,90,35,88,632.88	91.18
Total	1,07,42,12,02,180.14	56.62

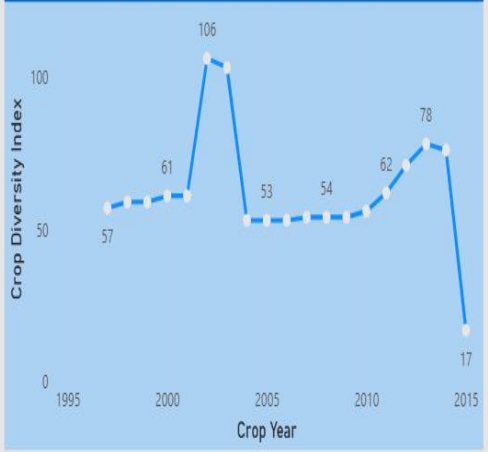
District Name	Total Production	Average Production Per Area
ALAPPUZHA	4,80,04,61,927.16	2,765.84
EAST GODAVARI	8,27,10,57,200.00	713.09
ERNAKULAM	5,02,16,49,325.77	2,603.48
KANNUR	9,78,34,32,072.12	3,248.43
KASARAGOD	7,73,22,16,929.42	4,130.40
KOLLAM	7,15,19,45,342.26	3,298.93
KOZHIKODE	15,28,07,39,862.95	5,094.17
MALAPPURAM	14,51,84,02,342.16	4,522.31
PALAKKAD	6,36,93,81,578.71	1,727.84
THIRUVANANTHAPURAM	10,02,27,14,452.33	4,518.80
Total	1,03,16,32,35,406.58	2,145.42

Microsoft Elevate

State Name

- Andaman and Nicobar Islands
- Andhra Pradesh
- Arunachal Pradesh
- Assam
- Bihar
- Chandigarh
- Chhattisgarh
- Dadra and Nagar Haveli
- Goa
- Gujarat
- Haryana

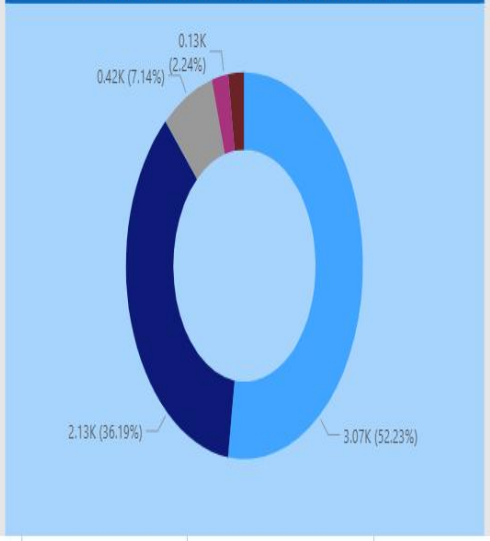
Crop Diversity Index by Crop Year



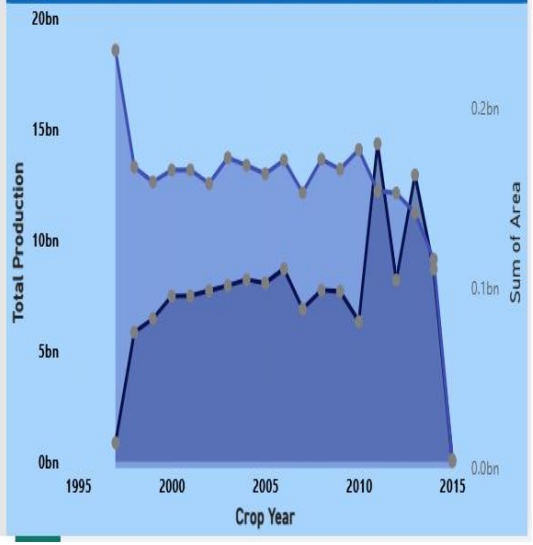
Total Production by State Name and Crop

Coconut Cotton(lint) Jute Maize Potato Rice Sugarcane Wheat

Average Production Per Area by State Name



Total Production and Sum of Area by Crop Year



Crop

- Coconut
- Cotton(lint)
- Potato
- Rice
- Sugarcane
- Wheat

1.25bn
Max of Production

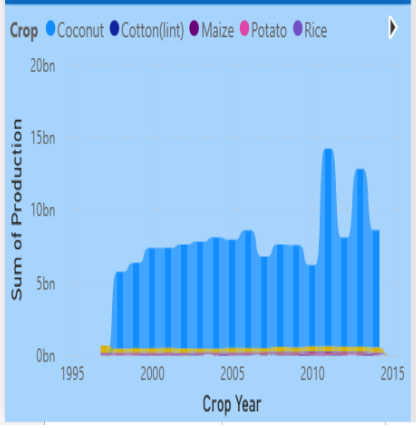
2K
Highest Production Year

47.79
Production Efficiency

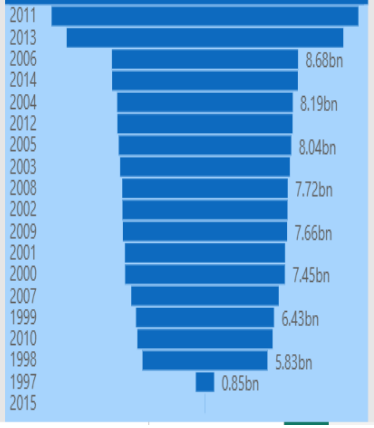
646
Total Districts

State Name	Banana	Coconut	Cotton(lint)	Jute	Maize	Potato	Rice
Andaman and Nicobar Islands	97,424.65	71,77,90,000.00			367.62		2,23,006.06
Andhra Pradesh	1,50,41,631.00	16,80,60,02,260.00	1,78,82,076.00		1,85,27,379.00	2,16,449.00	14,37,42,455.00
Arunachal Pradesh					10,58,829.00	5,78,571.00	27,09,476.60
Assam	1,17,24,386.00	1,95,44,08,105.00	12,341.00	1,17,89,532.00	3,89,382.00	1,10,21,521.00	7,18,01,090.00
Bihar	28,84,412.00		488.00	2,03,00,657.00	3,22,32,829.00	1,40,35,519.00	9,20,22,922.00
Chandigarh					4,225.00	13,795.00	3,853.00
Chhattisgarh	2,94,156.00	0.00	2,181.00	0.00	23,51,292.00	8,74,066.00	8,14,98,618.00
Dadra and Nagar Haveli	4,808.00	1,17,288.00			1,100.00		4,54,865.00
Total	14,61,32,679.64	1,29,98,16,29,216.29	29,70,00,015.70	18,15,58,177.90	27,33,41,803.57	42,48,26,343.86	1,60,54,70,383.37

Crop Performance Over Time



Crop Production Funnel



State Name and Season

Autumn Kharif Rabi Summer Whole Year Winter

CONCLUSION:

- This project presents a comprehensive analysis of crop production data across different states, crops, and seasons in India. By examining the dataset and organizing it into meaningful visual insights, the project helps in understanding how agricultural production varies across regions and over different years.
- Through the analysis, patterns in crop cultivation area, seasonal production, and regional contributions become more visible. The dashboard enables a clear comparison between different crops and states, making it easier to identify major production contributors and observe changes in agricultural output over time.
- The project also highlights the importance of data analytics in transforming large datasets into meaningful insights. By presenting agricultural data in a structured and visual format using Microsoft Power BI, the analysis becomes easier to interpret and more useful for understanding production trends.
- Overall, this project demonstrates how analytical tools and visualization techniques can be used to study agricultural data effectively and gain valuable insights into crop production patterns in India.

FUTURE SCOPE:

- The current system focuses on analyzing crop production data to understand agricultural trends across different regions and seasons. In the future, the system can be further enhanced by integrating additional agricultural datasets such as rainfall data, soil quality, and climate conditions. Including these factors can provide deeper insights into the reasons behind variations in crop production.
- Another possible improvement is the integration of predictive analytics techniques to forecast future crop production based on historical data and environmental conditions. This would help in identifying possible production trends and supporting better planning in the agricultural sector.
- Furthermore, the dashboard can be enhanced with advanced analytical features and real-time data updates, enabling more dynamic monitoring of agricultural performance. Incorporating advanced data analytics techniques and improved visualization methods can make the system more powerful and useful for large-scale agricultural analysis.

REFERENCES:

•**Arjun S., Anish Joshi, H. Pooja Das, Amutha R.**
Big Data Analytics for Agricultural Development in India.
International Journal of Engineering Research & Technology (IJERT), 2016.

•**Astha Tripathi, Mahesh G., Avneet Singh Charabra, Divyam Bahl.**
Forecasting Agricultural Crop Profitability and Yield in India.
International Journal for Research in Applied Science & Engineering Technology, 2024.

Github link : https://github.com/tejaswisrinivas24-gif/PowerBI-Crop_Production_Insights-India.git

Thank You